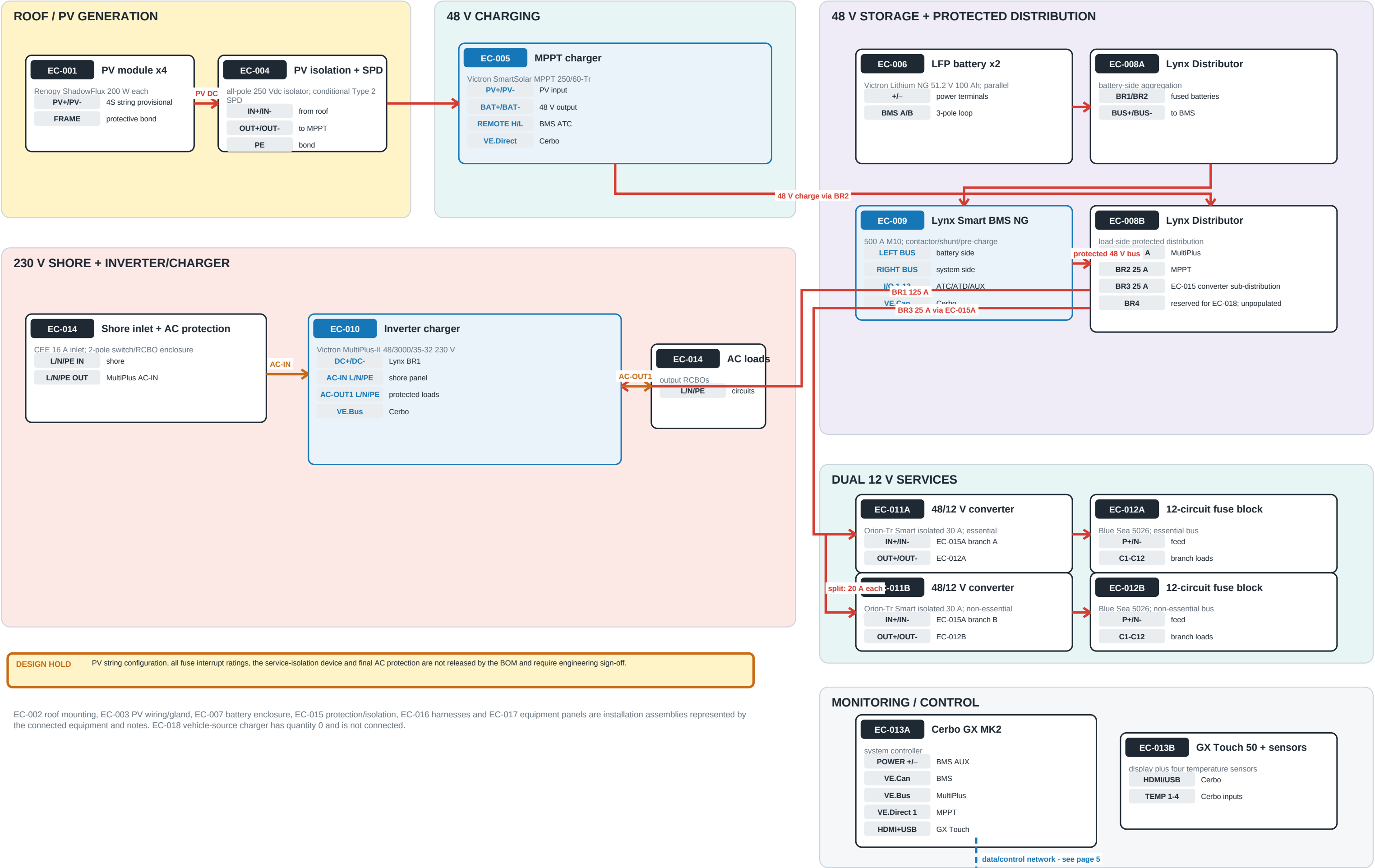


ELECTRICAL CORE - SYSTEM INTERCONNECTION OVERVIEW

DRAWING: EC-WD-001 | REV A

48 Vdc storage core | 800 W PV | 230 Vac inverter/charger | dual isolated 12 V service buses



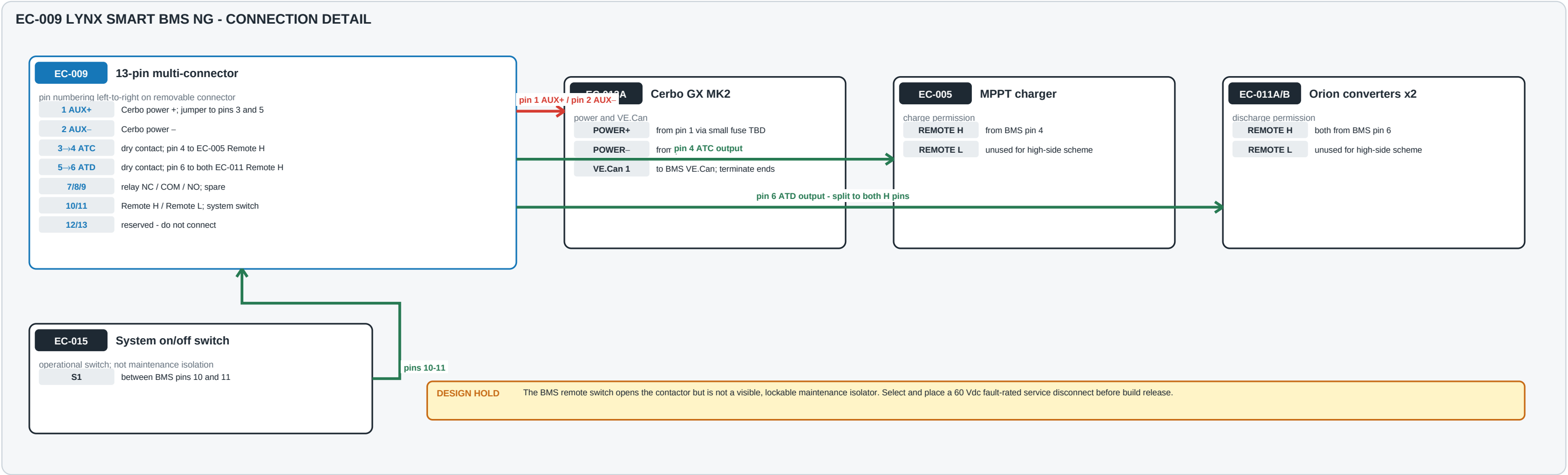
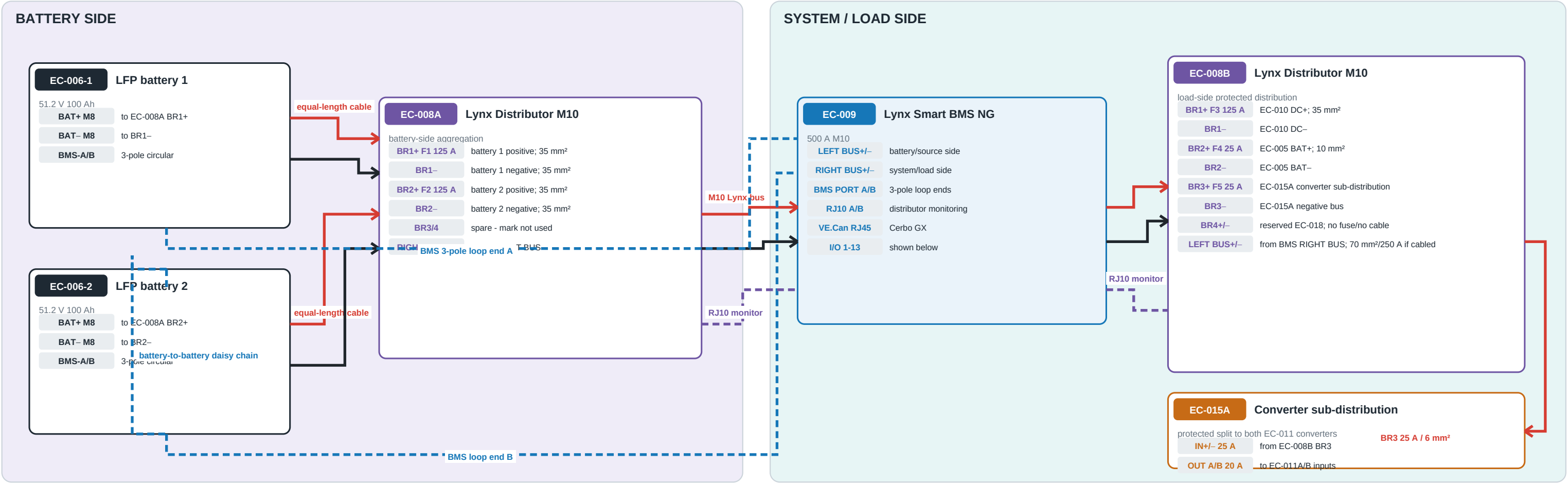
DESIGN HOLD

PV string configuration, all fuse interrupt ratings, the service-isolation device and final AC protection are not released by the BOM and require engineering sign-off.

EC-002 roof mounting, EC-003 PV wiring/gland, EC-007 battery enclosure, EC-015 protection/isolation, EC-016 harnesses and EC-017 equipment panels are installation assemblies represented by the connected equipment and notes. EC-018 vehicle-source charger has quantity 0 and is not connected.

48 V BATTERY, BMS AND LYNX DISTRIBUTION

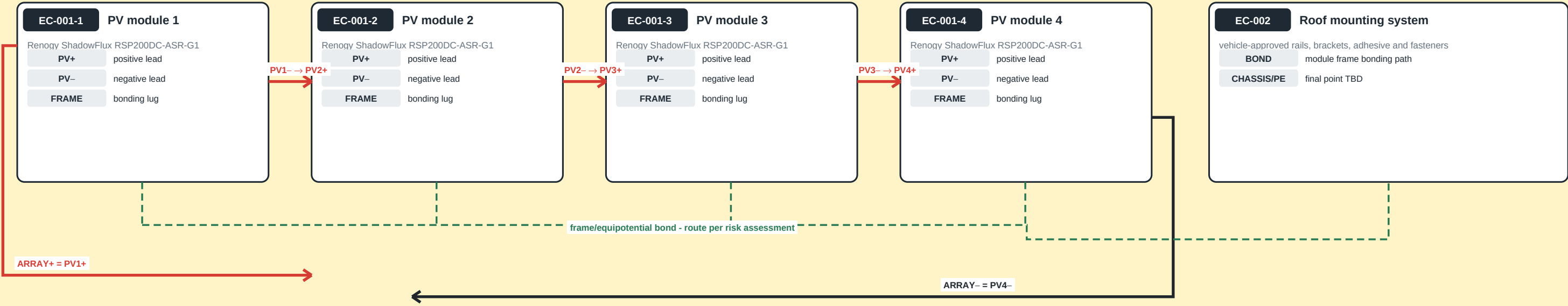
Power terminals, branch assignments, BMS communication loop and 13-pin I/O application



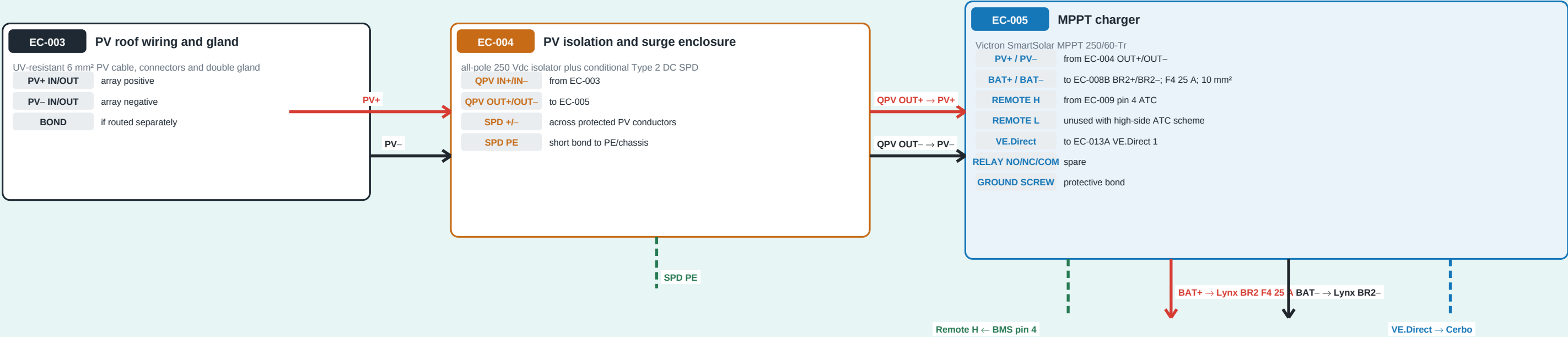
PV ARRAY, ISOLATION, SURGE PROTECTION AND MPPT

Component IDs and terminal-to-terminal connection detail

ROOF ARRAY - PROVISIONAL 4S STRING



TECHNICAL BAY - PV SWITCHING AND CHARGE CONTROL



ARRAY CHECKS / DESIGN HOLDS

- DESIGN HOLD**

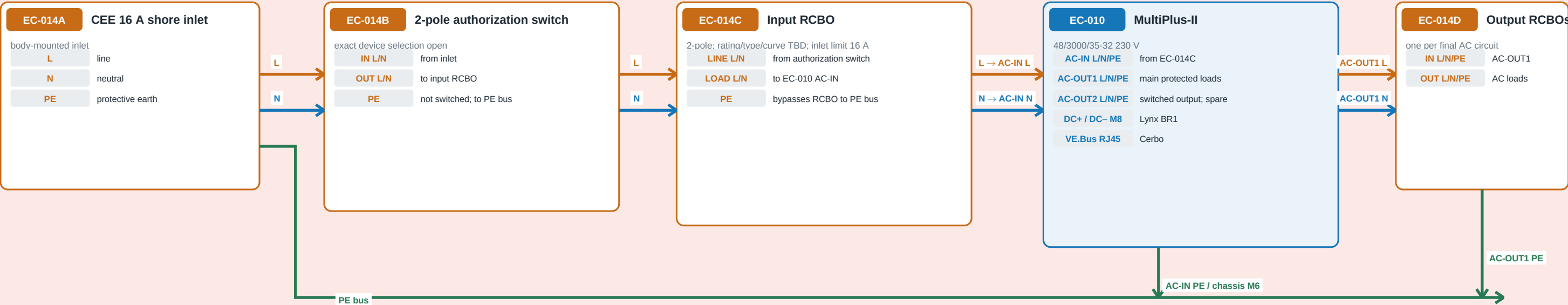
4S is a provisional connection inferred from four identical modules and one MPPT. Confirm cold-weather Voc remains below 250 V and perform shading/layout review before release.
- DESIGN HOLD**

SPD retention, frame bonding, cable routing and protective-device ratings require an installation-specific lightning/EMC/fault-risk assessment.

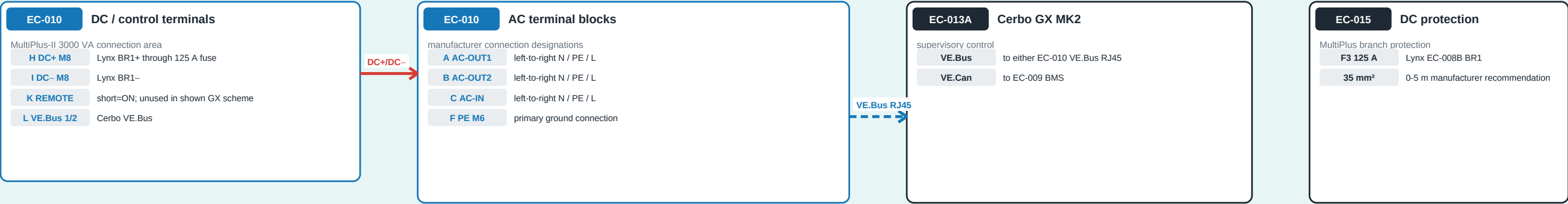
BOM basis: 4 × 200 W; module Voc 36.5 V, Vmp 31.3 V. Nominal 4S values: Voc 146 V, Vmp 125.2 V, power 800 W.

Functional pin designations; EC-014 protective-device models remain to be selected by a qualified designer

AC POWER PATH



EC-010 MULTIPLUS-II TERMINAL DETAIL



PROTECTIVE EARTH / DESIGN HOLDS

DESIGN HOLD

Select EC-014 authorization switch and RCBO arrangement for the jurisdiction, vehicle construction, prospective fault current and RCD compatibility. Do not infer ratings from this drawing alone.

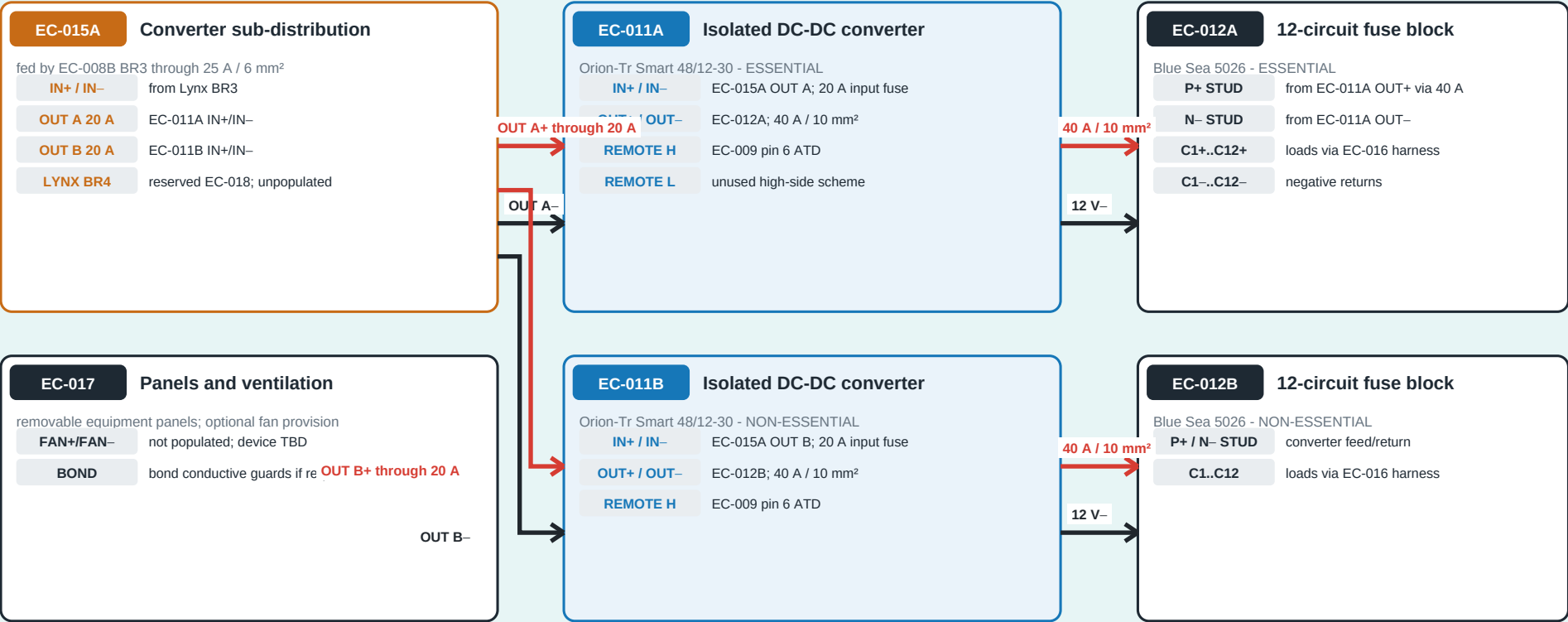
DESIGN HOLD

PE is never switched. Bond inlet PE, MultiPlus PE/chassis and output PE to the approved vehicle protective-bonding point. Neutral-earth behavior must follow the MultiPlus relay configuration and inspection regime.

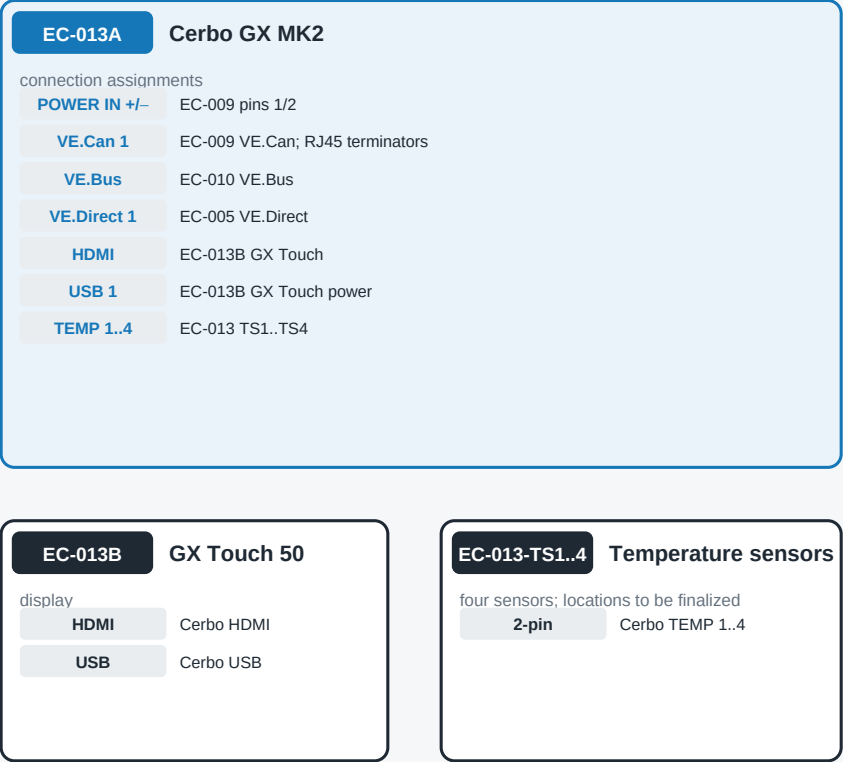
ISOLATED 12 V SERVICES AND MONITORING / DATA NETWORK

Separate essential and non-essential buses; Cerbo GX port assignments and BMS control signals

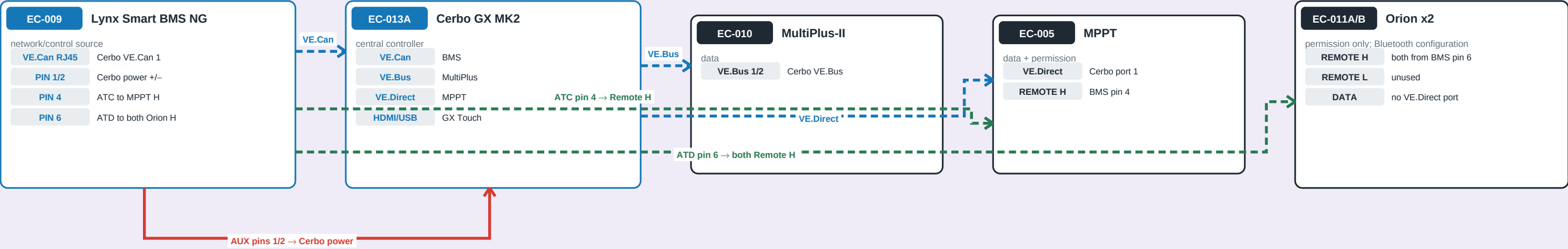
DUAL 48 V TO 12 V CONVERSION



CERBO GX / DISPLAY / SENSORS



NETWORK AND REMOTE-CONTROL CONNECTIONS



DESIGN HOLD The two 12 V negative buses are isolated from the 48 V negative by EC-011. Any chassis bond must be a deliberate single-point design decision; do not add casual cross-bonds.

DESIGN HOLD Assign every EC-012 branch a circuit ID, load name, cable size and fuse value. The 5026 limit is 100 A per block and 30 A per circuit; each converter limits available service current to 30 A.

TERMINAL-TO-TERMINAL CONNECTION SCHEDULE

Each connection is shown on pages 1-5; sizes marked TBD require measured route lengths and engineering calculations

CONN	FROM TERMINAL	TO TERMINAL	SERVICE	CABLE	PROTECTION / NOTE
C-001	EC-001-1 PV+	EC-003 ARRAY+	PV power	6 mm² PV	Provisional 4S end
C-002	EC-001-1 PV–	EC-001-2 PV+	PV power	module lead	Series link
C-003	EC-001-2 PV–	EC-001-3 PV+	PV power	module lead	Series link
C-004	EC-001-3 PV–	EC-001-4 PV+	PV power	module lead	Series link
C-005	EC-001-4 PV–	EC-003 ARRAY–	PV power	6 mm² PV	Provisional 4S end
C-006	EC-003 PV+/PV–	EC-004 QPV IN+/IN–	PV power	6 mm² PV	Through roof gland
C-007	EC-004 QPV OUT+/OUT–	EC-005 PV+/PV–	PV power	6 mm² PV	All-pole isolation
C-008	EC-004 SPD PE	Vehicle PE/bond point	Bond	TBD	Shortest practicable path
C-009	EC-006-1 BAT+/BAT–	EC-008A BR1+/BR1–	48 V power	35 mm² equal length	125 A provisional
C-010	EC-006-2 BAT+/BAT–	EC-008A BR2+/BR2–	48 V power	35 mm² equal length	125 A provisional
C-011	EC-006 BMS loop ends	EC-009 BMS ports A/B	Battery data	3-pole circular	Daisy-chain both batteries
C-012	EC-008A RIGHT BUS+/-	EC-009 LEFT BUS+/-	48 V bus	M10 Lynx link	Battery/source side
C-013	EC-009 RIGHT BUS+/-	EC-008B LEFT BUS+/-	48 V bus	M10 link or 70 mm²	250 A provisional; system/load side
C-014	EC-008B BR1+/-	EC-010 DC+/DC– M8	48 V power	35 mm² ≤5 m	125 A fuse
C-015	EC-008B BR2+/-	EC-005 BAT+/BAT–	48 V charge	10 mm²	25 A provisional; sized to installed 800 W array
C-016	EC-008B BR3+/-	EC-015A IN+/IN–	48 V power	6 mm²	25 A provisional
C-017	EC-015A OUT A/B +/-	EC-011A/B IN+/IN–	48 V power	2.5-6 mm² by run	20 A each; BR4 reserved for EC-018
C-018	EC-011A OUT+/OUT–	EC-012A P+/N–	12 V power	10 mm²	40 A provisional; essential
C-019	EC-011B OUT+/OUT–	EC-012B P+/N–	12 V power	10 mm²	40 A provisional; non-essential
C-020	EC-014 inlet L/N/PE	EC-014B IN L/N + PE bus	230 Vac	TBD approved cable	16 A inlet
C-021	EC-014C LOAD L/N + PE	EC-010 AC-IN L/N/PE	230 Vac	TBD approved cable	2-pole input protection
C-022	EC-010 AC-OUT1 L/N/PE	EC-014D output RCBOs	230 Vac	TBD approved cable	Final circuits TBD
C-023	EC-009 VE.Can	EC-013A VE.Can 1	Data	RJ45 VE.Can	Terminate first and last device
C-024	EC-010 VE.Bus	EC-013A VE.Bus	Data	RJ45 UTP	Use either MultiPlus VE.Bus socket
C-025	EC-005 VE.Direct	EC-013A VE.Direct 1	Data	VE.Direct cable	Maximum route per manufacturer
C-026	EC-009 pin 1/2 AUX+/-	EC-013A POWER+/-	Aux power	small fused pair	BMS AUX max 1.1 A
C-027	EC-009 pin 4 ATC	EC-005 Remote H	Control	small signal wire	Pin 3 jumpered from AUX+
C-028	EC-009 pin 6 ATD	EC-011A/B Remote H	Control	small signal wire	Pin 5 jumpered from AUX+
C-029	EC-009 pin 10/11	EC-015 system switch	Control	2-core	Operational contactor control
C-030	EC-013A HDMI + USB	EC-013B GX Touch	Display	integral display cable	Connect before powering Cerbo
C-031	EC-013 TS1..TS4	EC-013A TEMP1..TEMP4	Sensors	sensor leads	Locations TBD
C-032	EC-008A/B RJ10	EC-009 RJ10 A/B	Fuse monitoring	RJ10	One cable per distributor

DESIGN HOLD This schedule is a connection map, not a released cable schedule. Replace every TBD with conductor material, cross-section, insulation class, route length, voltage-drop result, fuse type/rating/interrupt capacity and termination specification before construction.

BOM CROSS-REFERENCE, DESIGN HOLDS AND SOURCE MANUALS

Every electrical-core-bom.csv row is represented; quantity-zero item is explicitly excluded

A. BOM CROSS-REFERENCE

ID	SUBSYSTEM	COMPONENT NAME	QTY	CANDIDATE / DESCRIPTION	DIAGRAM STATUS
EC-001	solar	PV module	4	Renogy ShadowFlux 200 W	Connected; provisional 4S
EC-002	solar	Roof mounting system	1	Rails/brackets/adhesive/fasteners	Mechanical + bond design hold
EC-003	solar	PV roof wiring and gland	1	6 mm² PV cable/connectors/double gland	Connected
EC-004	solar	PV isolation and surge enclosure	1	250 Vdc all-pole isolator + conditional SPD	Selection/risk hold
EC-005	solar	MPPT charger	1	Victron SmartSolar MPPT 250/60-Tr	Connected
EC-006	storage	LFP battery	2	Victron Lithium NG 51.2 V 100 Ah	Parallel power + daisy-chain BMS
EC-007	storage	Battery enclosure and restraint	1	Restraint/thermal barriers/service cradle	No electrical pins; bond if conductive
EC-008	distribution	Lynx Distributor	2	Victron Lynx Distributor M10	A=battery side; B=load side
EC-009	storage	Battery management system	1	Lynx Smart BMS NG 500A M10	Connected; pin map pages 2/5
EC-010	ac power	Inverter charger	1	MultiPlus-II 48/3000/35-32 230 V	Connected
EC-011	12v conversion	Isolated DC-DC converter	2	Orion-Tr Smart 48/12-30 isolated	A=essential; B=non-essential
EC-012	12v distribution	12-circuit fuse block	2	Blue Sea 5026 ST Blade	Branch assignments TBD
EC-013	monitoring	System controller display and sensors	1 set	Cerbo GX MK2 + GX Touch 50 + 4 sensors	Connected; sensor locations TBD
EC-014	ac power	Shore inlet and AC protection	1 set	CEE 16 A + 2-pole switch/RCBOS	Device selection/design hold
EC-015	distribution	DC protection and isolation	1 set	Fuses/holders/service disconnect/guards	Partial ratings shown; fault study hold
EC-016	distribution	Power and signal harnesses	1 set	Conductors/lugs/supports/data/labels	Measured cable schedule required
EC-017	mechanical	Equipment panels and ventilation	1 set	Panels/rails/guards/grilles/fan provision	Fan not selected
EC-018	vehicle interface	Vehicle-source charger	0	Renault-approved interface and charger	DEFERRED - NO CONNECTION

B. RELEASE-BLOCKING DESIGN HOLDS

- H1 - Confirm PV 4S topology using cold-weather Voc, shading and roof-layout calculations.
- H2 - Complete 48 V battery fault-current study; select battery branch fuses, fuse class and interrupt ratings.
- H3 - Select a visible, lockable 60 Vdc maintenance isolator; the BMS remote switch is only operational control.
- H4 - Measure every route and release the EC-016 cable schedule including voltage drop, derating and terminations.
- H5 - Qualified AC designer to release EC-014 inlet, authorization, RCBO/RCD, PE and neutral-earth arrangement.
- H6 - Assign all 24 EC-012 branch circuits and decide whether either isolated 12 V negative bus bonds to chassis.
- H7 - Verify crash restraint, thermal management, IP protection, ventilation and service clearances.
- H8 - EC-018 remains disconnected until Renault approval evidence and a compatible charger/interface exist.

C. PRIMARY TECHNICAL SOURCES

- BOM: electrical-core-bom.csv
- Victron Lynx Smart BMS NG manual: https://www.victronenergy.com/media/pg/Lynx_Smart_BMS_NG/en/index-en.html
- Victron Lynx Distributor manual: https://www.victronenergy.com/media/pg/Lynx_Distributor/en/index-en.html
- Victron Lithium NG 51.2 V manual: https://www.victronenergy.com/media/pg/Lithium_NG_battery_51%2C2_V/en/index-en.html
- Victron SmartSolar 150/60-250/70 manual: https://www.victronenergy.com/media/pg/Manual_SmartSolar_MPPT_150-60_up_to_250-70/en/index-en.html
- Victron MultiPlus-II 230 V manual: https://www.victronenergy.com/media/pg/MultiPlus-II_230V/en/index-en.html
- Victron Orion-Tr Smart isolated manual: https://www.victronenergy.com/media/pg/Orion-Tr_Smart_DC-DC_Charger_-_Isolated/en/index-en.html
- Victron Cerbo GX manual: https://www.victronenergy.com/media/pg/Cerbo_GX/en/index-en.html
- Blue Sea Systems 5026: <https://www.bluesea.com/products/5026/ST%20Blade%20Fuse%20Block%20-%2012%20Circuits%20with%20Negative%20Bus%20and%20Cover>